Experiment No. 12

Demonstration of Virtualization Using a Type-2 Hypervisor

AIM

To demonstrate virtualization by installing and using VMware Workstation (a Type-2 hypervisor) on a Windows host, and creating, configuring, and installing an Ubuntu Linux virtual machine (guest operating system)

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SOFTWARE AND HARDWARE REQUIREMENTS

Host OS	Windows 10/11
Hypervisor	VMware Workstation Pro 17
Guest OS	Ubuntu 64-bit desktop ISO image
Recommended resources	CPU virtualization enabled; 2 virtual processors, 4 GB RAM if available, and at least 20 GB virtual disk

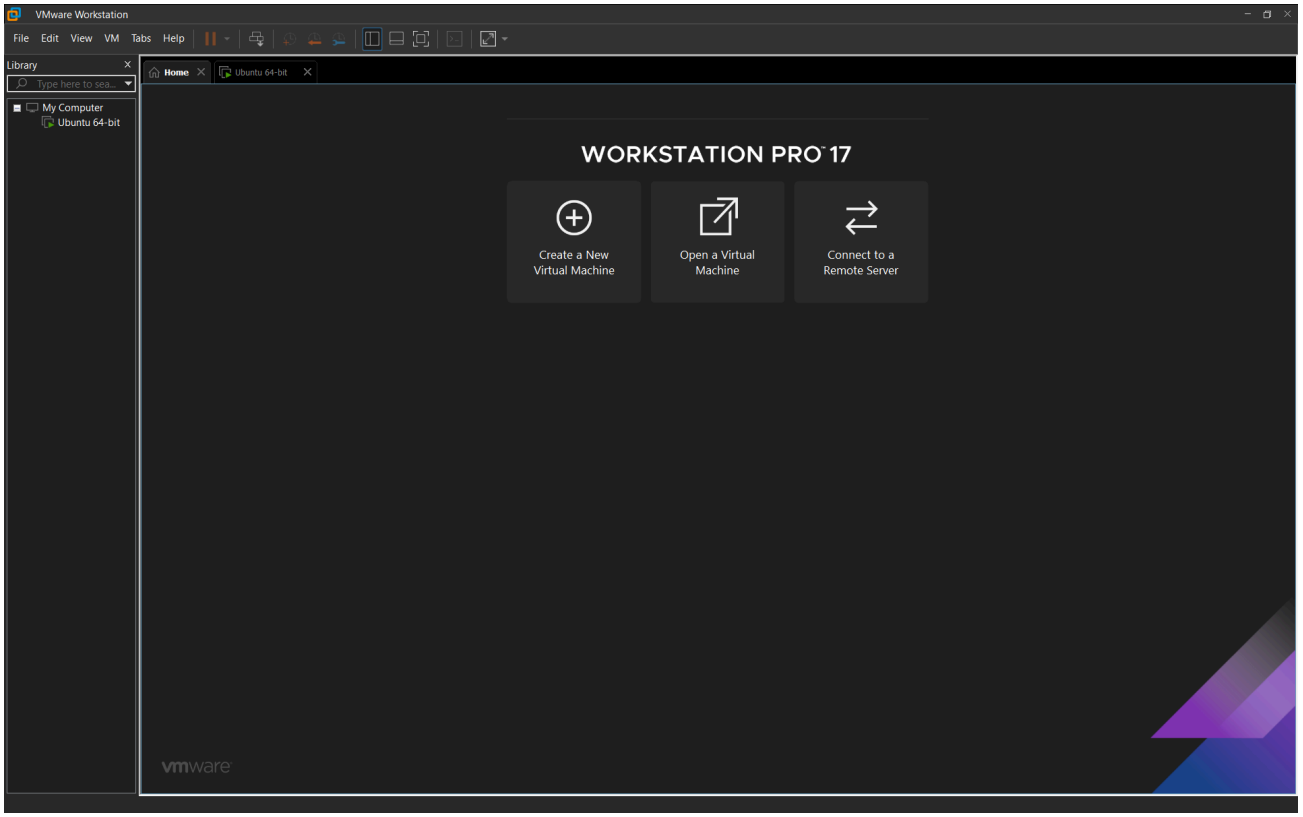
BRIEF THEORY

A Type-2 (hosted) hypervisor runs as an application on an existing host operating system and creates isolated virtual machines. VMware Workstation allows Ubuntu Linux to run as a guest on a Windows host without replacing the host OS.

PROCEDURE WITH SCREENSHOTS

Step 1. Open VMware Workstation Pro.

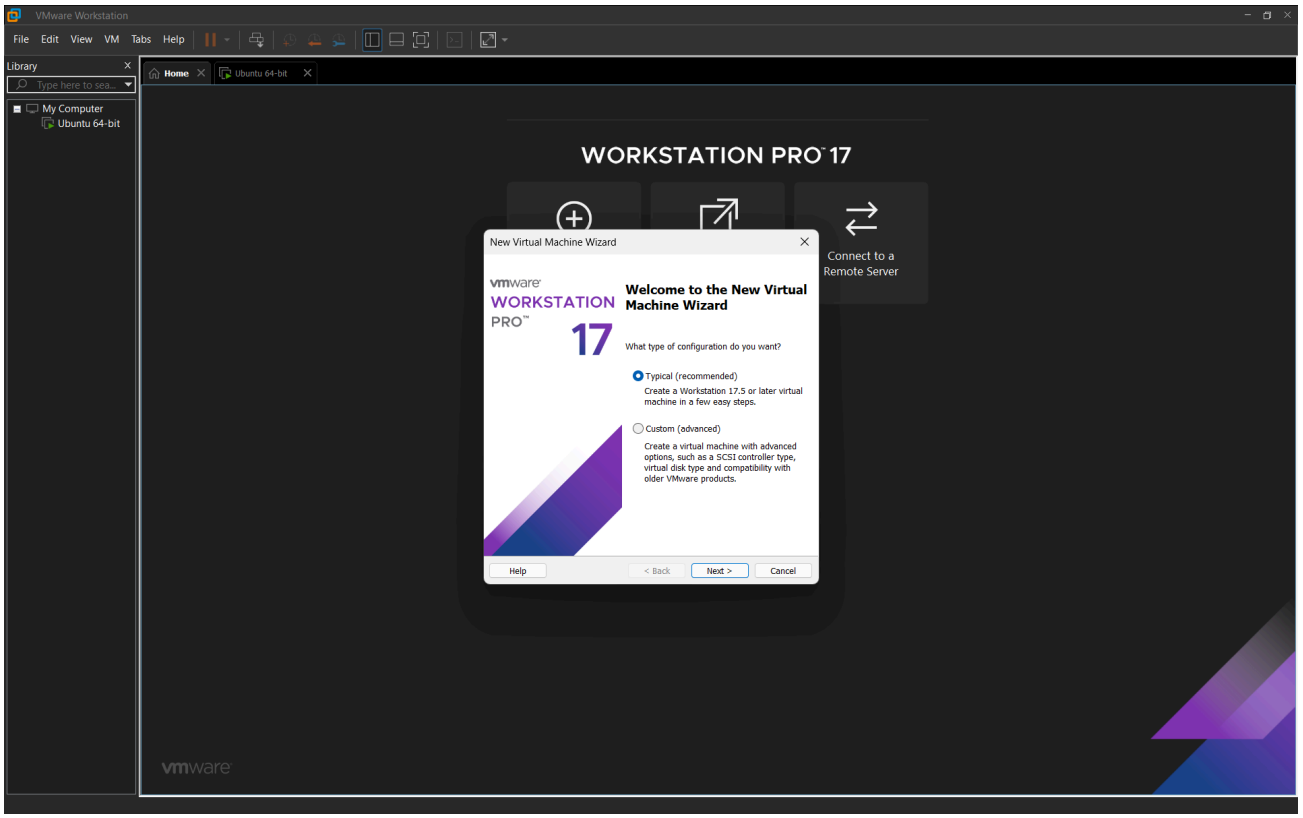
Launch VMware Workstation on the Windows host computer and select Create a New Virtual Machine.



Snap 1 – VMware Workstation home page

Step 2. Select the configuration type.

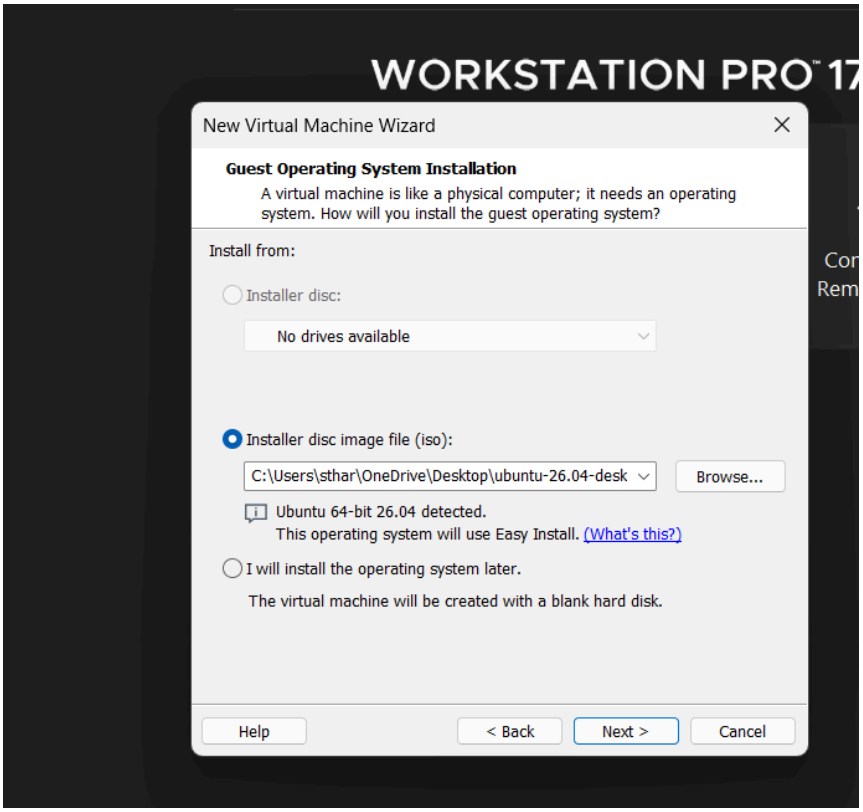
In the New Virtual Machine Wizard, select Typical (recommended), then click Next.



Snap 2 – Welcome to the New Virtual Machine Wizard

Step 3. Choose the Ubuntu installation image.

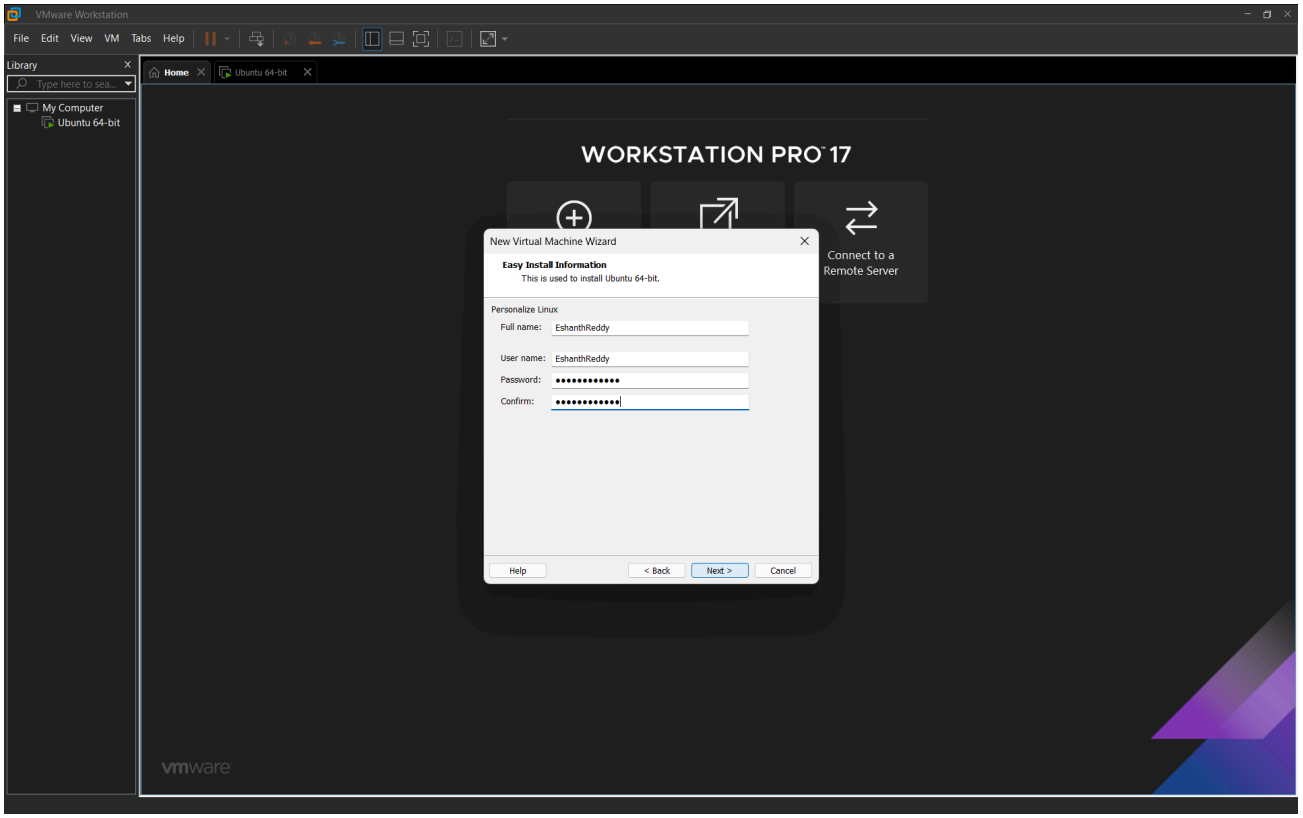
Under Guest Operating System Installation, select Installer disc image file (iso), browse to the downloaded Ubuntu desktop ISO, verify that Ubuntu 64-bit is detected, and click Next.



Snap 3 – Guest operating system / ISO selection

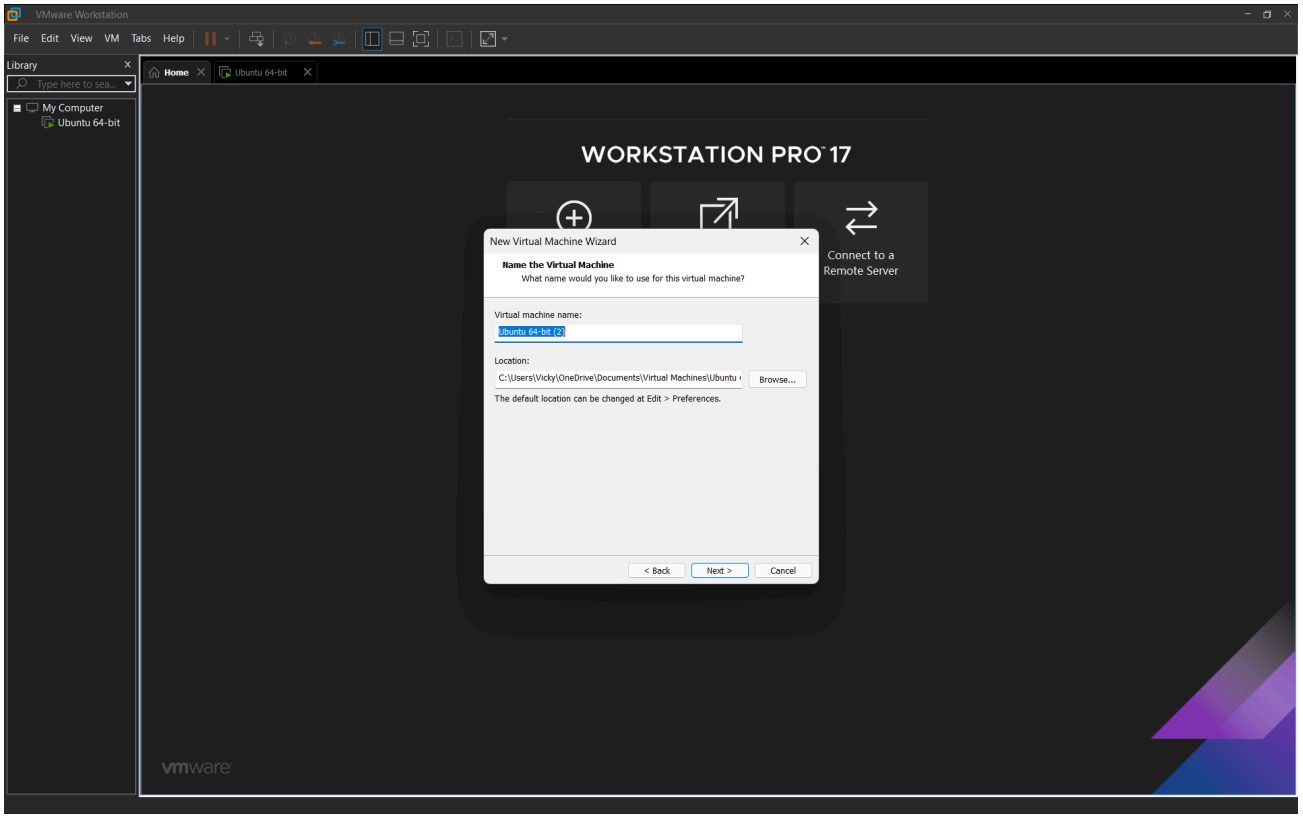
Step 4. Personalize the Linux installation.

If the Easy Install page appears, enter the requested full name, username, and password, then click Next. Keep the password secure.



Step 5. Name the virtual machine.

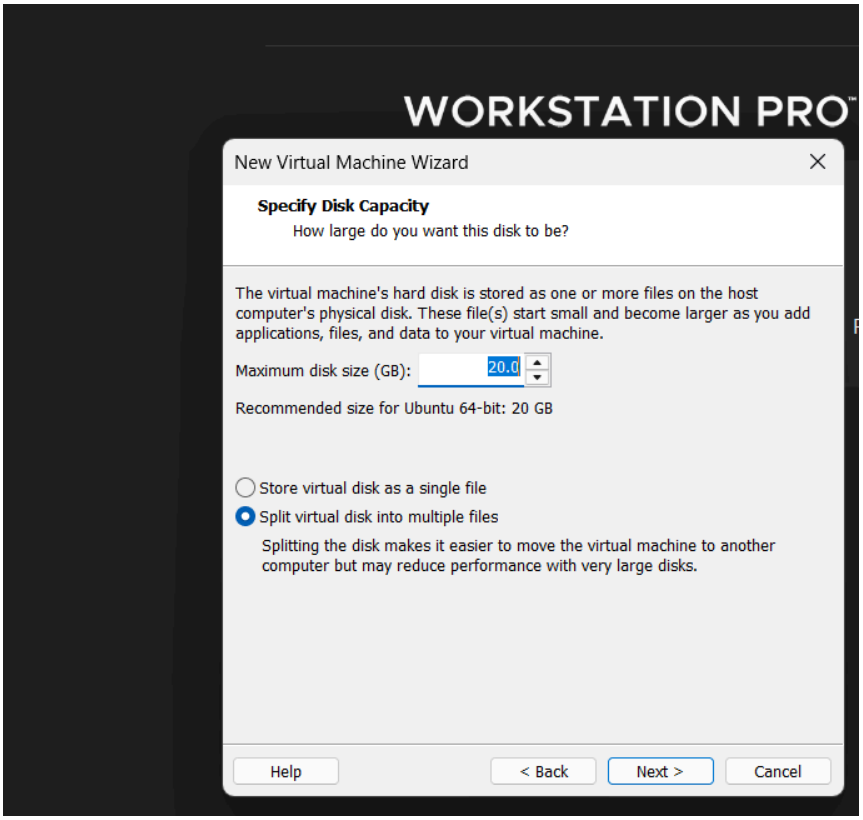
Enter a virtual machine name such as Ubuntu, choose a storage location on the Windows computer, and click Next. Prefer a local folder with sufficient free space rather than a cloud-synchronized folder.



Snap 5 – Virtual machine name and location

Step 6. Specify the disk capacity.

Set the maximum virtual disk size to 20 GB as shown, choose Split virtual disk into multiple files, and click Next. More disk space may be beneficial for regular Ubuntu desktop use.



Snap 6 – Virtual disk capacity and storage option

Step 7. Customize virtual hardware.

On the Ready to Create Virtual Machine screen, click Customize Hardware. Allocate suitable memory and processors (for example, 4 GB RAM and 2 processor cores if the host permits), confirm network settings, and verify the ISO is attached. Close the hardware settings and click Finish.

Step 8. Start the virtual machine.

Power on the newly created VM. The Ubuntu installer starts from the configured ISO; wait for the Ubuntu installation screen.

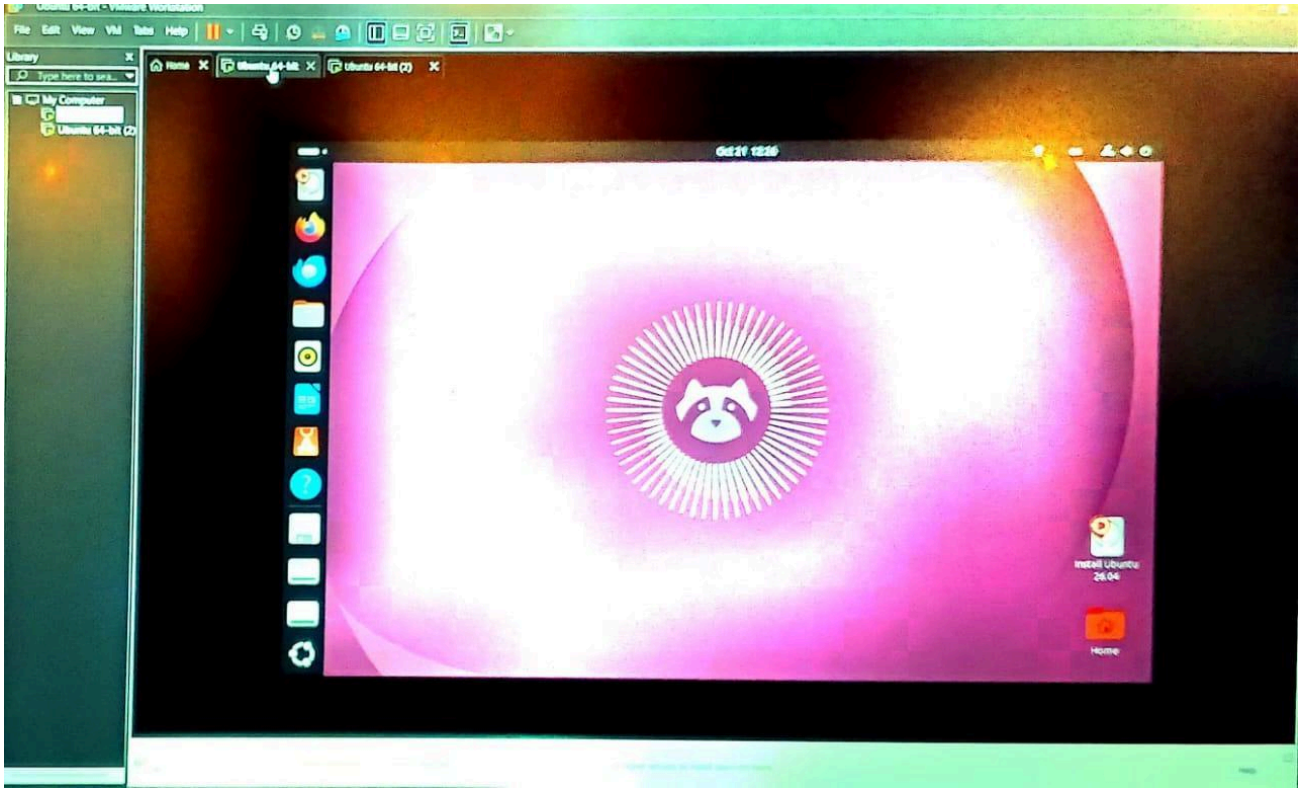
Step 9. Install Ubuntu Linux.

Follow the on-screen installer prompts for language, keyboard, installation type, user account, and time zone. Select the VM's virtual disk as the installation target, then begin installation. These changes are made inside the VM.

Snap 9 – Ubuntu installation in progress

Step 10. Complete setup and verify the guest OS.

After installation, restart the virtual machine if prompted, sign in to Ubuntu, and verify that the Ubuntu desktop loads correctly.



Snap 10 – Ubuntu desktop / successful installation

OBSERVATION

VMware Workstation operates as a Type-2 hypervisor on the Windows host. A separate Ubuntu 64-bit virtual machine is created with a virtual disk and configurable CPU, memory, and network resources.

RESULT

The Type-2 hypervisor was configured and the procedure for creating and installing an Ubuntu Linux guest virtual machine was demonstrated. Confirm successful installation after adding the final Ubuntu desktop screenshot.

PRECAUTIONS

Enable hardware virtualization in BIOS/UEFI if required; ensure adequate available host RAM and storage; attach a valid Ubuntu ISO; and perform installation only within the intended virtual disk.

Experiment No. 13

Creating a Virtual Machine with 1 CPU, 2 GB RAM and 15 GB Storage Disk Using a Type-2 Virtualization Software

AIM

To create a virtual machine with 1 CPU, 2 GB RAM and a 15 GB storage disk using a Type-2 virtualization software (VMware Workstation).

SOFTWARE AND HARDWARE REQUIREMENTS

Host OS	Windows 10/11
Hypervisor	VMware Workstation Pro 17
Guest OS	Ubuntu 64-bit desktop ISO image
VM configuration	1 processor, 2 GB (2048 MB) RAM, 15 GB virtual disk

PROCEDURE WITH SCREENSHOTS

Step 1. Start the New Virtual Machine Wizard.

Open VMware Workstation and click Create a New Virtual Machine. Select Typical (recommended) and click Next.

Step 2. Select the installer image.

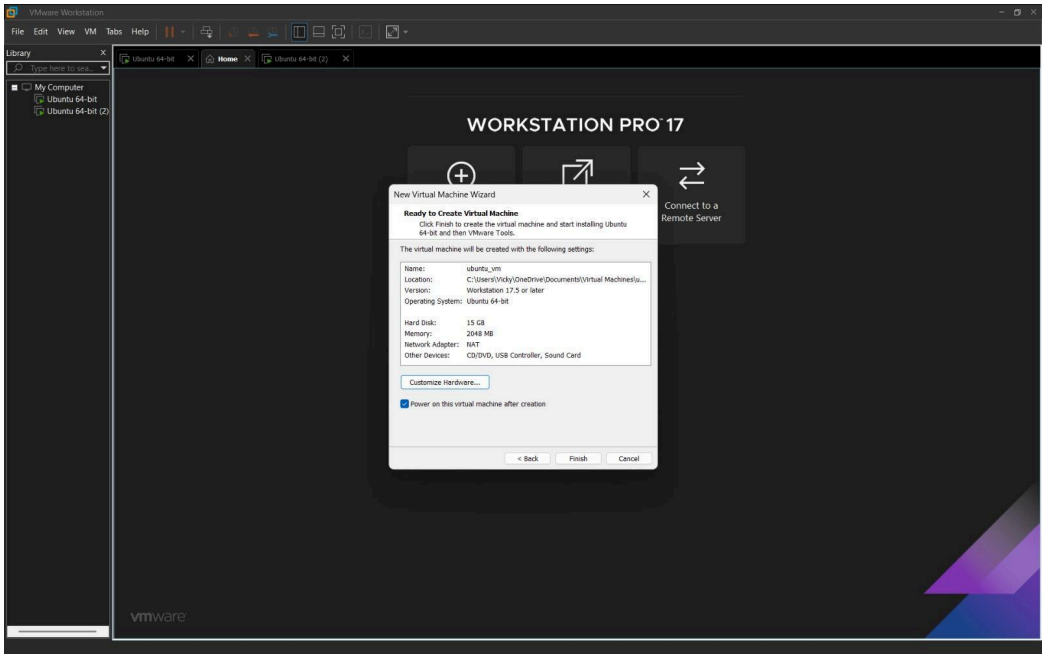
Select Installer disc image file (iso), click Browse and select the Ubuntu/Linux ISO file, then click Next.

Step 3. Name the VM and set the disk size.

Enter the virtual machine name as ubuntu_vm and select the location where the VM is to be saved. Set the maximum disk size to 15 GB and select Store virtual disk as a single file.

Step 4. Review the settings on the Ready to Create screen.

The summary shows the name ubuntu_vm, operating system Ubuntu 64-bit, hard disk 15 GB, memory 2048 MB and network adapter NAT. Click Customize Hardware to set the CPU and RAM (2 GB memory, 1 processor), then click Close and Finish.



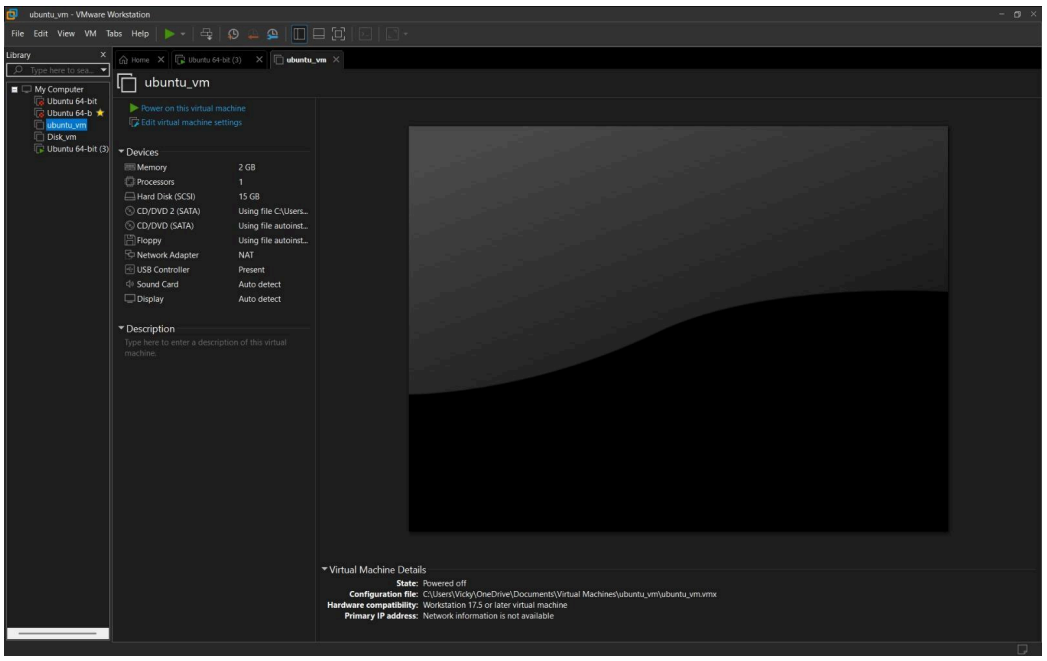
Snap 1 – Ready to Create Virtual Machine (15 GB disk, 2048 MB memory)

Step 5. Power on and install the guest OS.

Click Power on this virtual machine. The VM boots from the ISO and starts the Ubuntu installation; complete the installation and restart the VM.

Step 6. Verify the final configuration.

The ubuntu_vm summary page lists Memory 2 GB, Processors 1 and Hard Disk (SCSI) 15 GB, with the network adapter set to NAT.



Snap 2 – ubuntu_vm summary: 2 GB memory, 1 processor, 15 GB hard disk

OBSERVATION

The virtual machine ubuntu_vm was created with 1 virtual processor, 2 GB of RAM and a 15 GB virtual hard disk, and a NAT network adapter.

RESULT

A virtual machine with 1 CPU, 2 GB RAM and a 15 GB storage disk was successfully created using VMware Workstation Type-2 virtualization software.

PRECAUTIONS

Ensure the host has enough free RAM and disk space; attach a valid Ubuntu ISO; and enable hardware virtualization in BIOS/UEFI if required.

Experiment No. 14

Creating a Virtual Hard Disk and Allocating Storage Using VMware Workstation

AIM

To create a virtual hard disk and allocate storage to a virtual machine using VMware Workstation.

SOFTWARE AND HARDWARE REQUIREMENTS

Host OS	Windows 10/11
Hypervisor	VMware Workstation Pro 17
Virtual machine	Ubuntu 64-bit VM (powered off)

BRIEF THEORY

A virtual hard disk is a file (.vmdk) on the host computer that appears to the guest operating system as a physical disk. Storage can be allocated to a VM by adding a new virtual disk in the VM hardware settings.

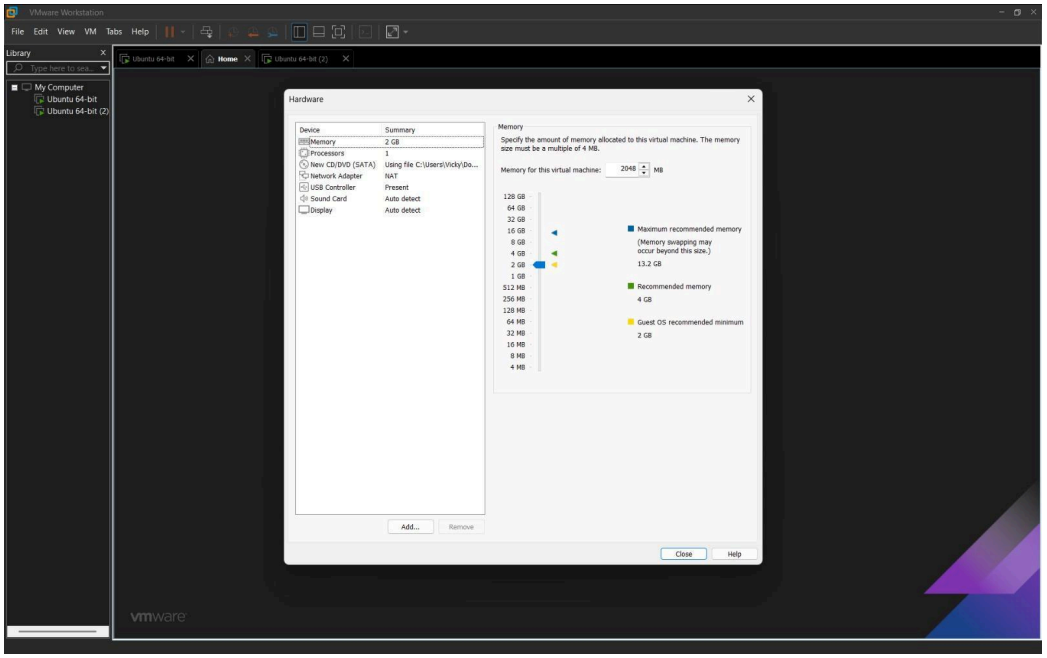
PROCEDURE WITH SCREENSHOTS

Step 1. Open the virtual machine settings.

Power off the virtual machine, select it in the Library and click Edit virtual machine settings.

Step 2. Open the Hardware window.

The Hardware window lists the devices of the VM (Memory, Processors, CD/DVD, Network Adapter, USB Controller, Sound Card and Display). Click Add… to add a new device.



Snap 1 – Hardware window with the Add… button for adding a new device

Step 3. Add a hard disk.

In the Add Hardware Wizard select Hard Disk and click Next. Choose the recommended disk type and click Next.

Step 4. Create a new virtual disk.

Select Create a new virtual disk and click Next.

Step 5. Allocate the storage.

Set the maximum disk size (for example 15 GB). Choose whether to allocate all disk space now, and select Store virtual disk as a single file or Split virtual disk into multiple files. Click Next.

Step 6. Name the disk file and finish.

Specify the disk file name and location, click Finish, and then click OK to save the settings.

Step 7. Verify the new disk.

Power on the VM and confirm that the newly added virtual hard disk is listed in the VM settings and is available to the guest operating system.

OBSERVATION

A new virtual hard disk was created and the storage was allocated to the virtual machine through the VMware Workstation hardware settings.

RESULT

A virtual hard disk was created and storage was successfully allocated using VMware Workstation.

PRECAUTIONS

Ensure enough free space on the host drive; power off the VM before adding hardware; and choose the disk size carefully, as it cannot easily be reduced later.

Experiment No. 15

Creating a Snapshot of a Virtual Machine and Testing it by Loading the Previous Version

AIM

To create a snapshot of a virtual machine and test it by restoring (loading) the previous version of the virtual machine.

SOFTWARE AND HARDWARE REQUIREMENTS

Host OS	Windows 10/11
Hypervisor	VMware Workstation Pro 17
Virtual machine	ubuntu_vm (1 CPU, 2 GB RAM, 15 GB disk)

BRIEF THEORY

A snapshot preserves the state of a virtual machine at a particular point in time so that the VM can be returned to the same state later. It is useful before installing software or making risky configuration changes.

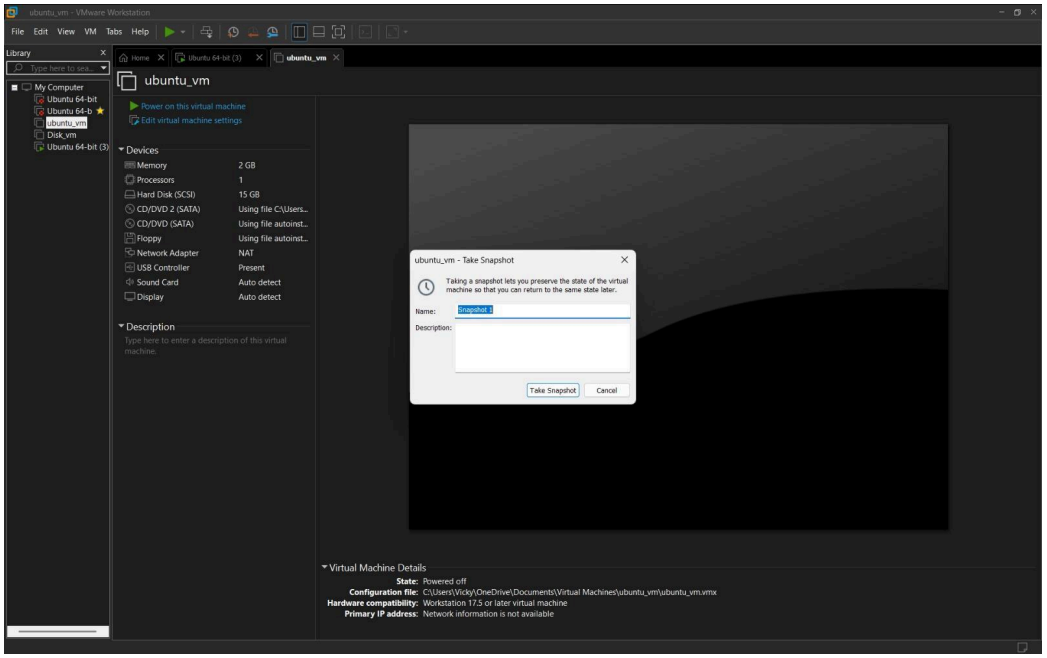
PROCEDURE WITH SCREENSHOTS

Step 1. Select the virtual machine.

Open VMware Workstation and select ubuntu_vm in the Library.

Step 2. Take the snapshot.

From the VM menu choose Snapshot → Take Snapshot (or click the Take Snapshot toolbar button). Enter a name for the snapshot, optionally add a description, and click Take Snapshot.



Snap 1 – Take Snapshot dialog for ubuntu_vm (Snapshot 1)

Step 3. Change the state of the VM.

Make a change inside the VM (for example create a file or install a package) so that it differs from the snapshot.

Step 4. Load the previous version.

Open VM → Snapshot → Snapshot Manager, select Snapshot 1 and click Go To (or use Revert to Snapshot).

Step 5. Test the restored VM.

Power on the VM and confirm that the change made in Step 3 is no longer present, i.e. the VM has returned to its state at the time of the snapshot.

OBSERVATION

A snapshot named Snapshot 1 was taken of ubuntu_vm. After restoring it, the virtual machine returned to the saved state.

RESULT

A snapshot of the virtual machine was created and tested by loading the previous version of the VM.

PRECAUTIONS

Take snapshots only when sufficient disk space is available; do not keep too many snapshots, as they slow the VM and use storage; changes made after the snapshot are lost when it is restored.

Experiment No. 16

Cloning a Virtual Machine and Testing it by Loading the Cloned VM

AIM

To create a clone of a virtual machine and test it by loading the cloned virtual machine.

SOFTWARE AND HARDWARE REQUIREMENTS

Host OS	Windows 10/11
Hypervisor	VMware Workstation Pro 17
Source VM	ubuntu_vm (powered off)

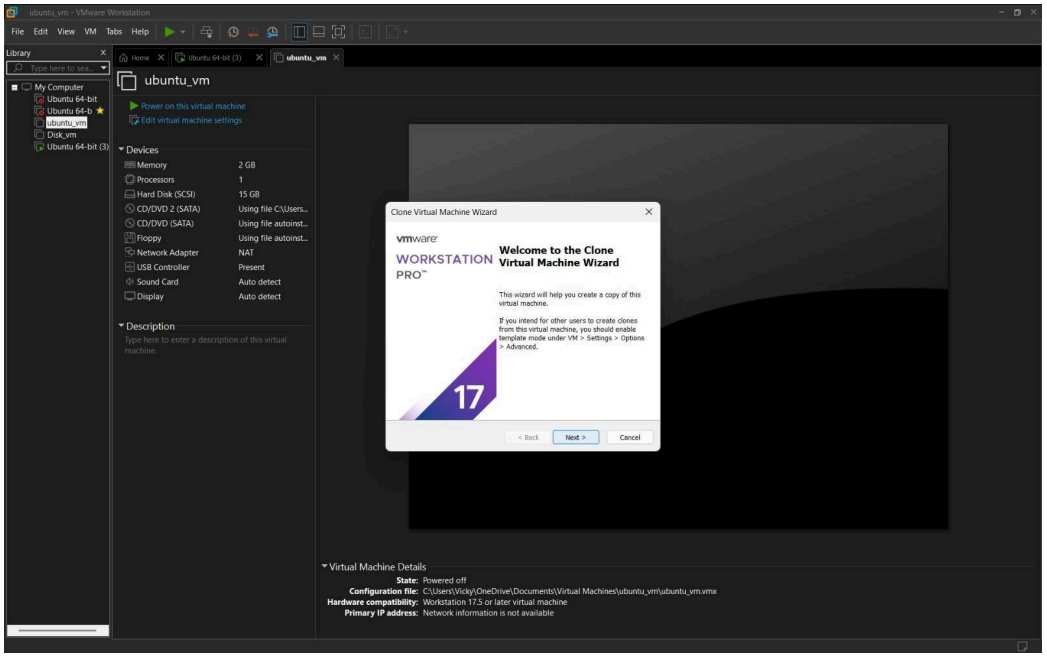
BRIEF THEORY

A clone is a copy of an existing virtual machine. A full clone is an independent copy, while a linked clone shares the virtual disk of the parent VM and takes less disk space.

PROCEDURE WITH SCREENSHOTS

Step 1. Start the Clone Virtual Machine Wizard.

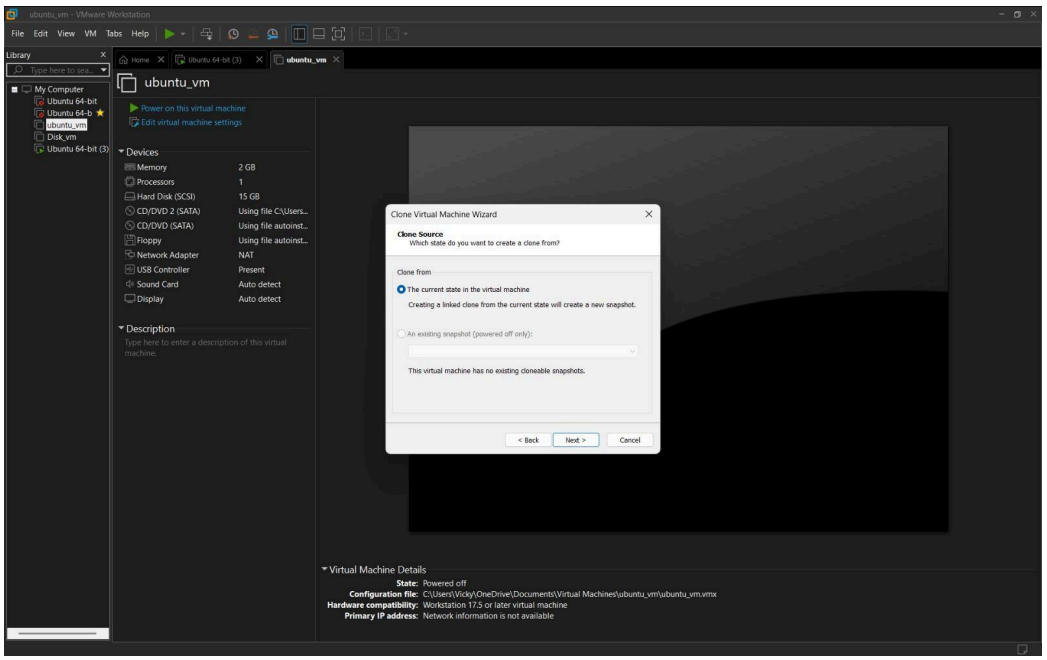
Make sure ubuntu_vm is powered off, then right-click it and choose Manage → Clone (or VM → Manage → Clone). The Welcome page of the Clone Virtual Machine Wizard opens; click Next.



Snap 1 – Welcome to the Clone Virtual Machine Wizard

Step 2. Choose the clone source.

Select The current state in the virtual machine (or an existing snapshot) and click Next.



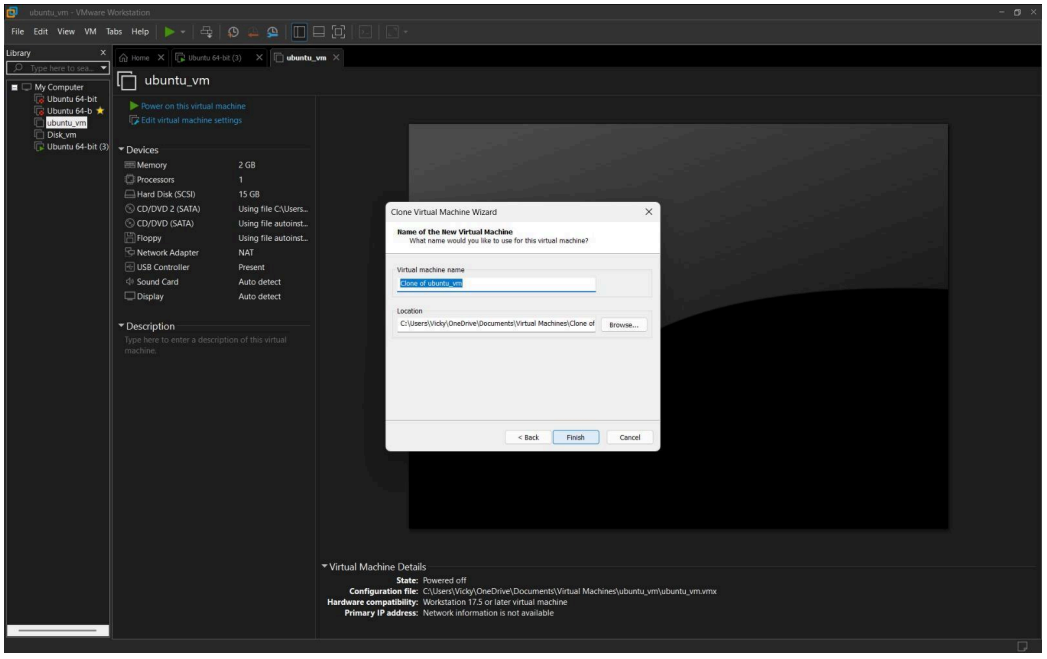
Snap 2 – Clone Source: the current state in the virtual machine

Step 3. Select the clone type.

Choose Create a linked clone or Create a full clone, and click Next.

Step 4. Name the cloned VM.

Enter the name of the new virtual machine (default: Clone of ubuntu_vm), choose its location and click Finish.



Snap 3 – Name and location of the new virtual machine

Step 5. Test the cloned VM.

When the cloning completes, click Close. The cloned VM appears in the Library; select it and click Power on this virtual machine. Confirm that it boots and works like the original ubuntu_vm.

OBSERVATION

A clone of ubuntu_vm named Clone of ubuntu_vm was created with the same configuration (1 CPU, 2 GB RAM, 15 GB disk) as the original.

RESULT

The virtual machine was cloned successfully and the cloned VM was loaded and tested.

PRECAUTIONS

Power off the source VM before cloning; keep enough free disk space for the clone; and do not delete the parent VM if a linked clone is used.